1. จุติณัฏฐ์ รัตนะมงคลกุล 6609650228

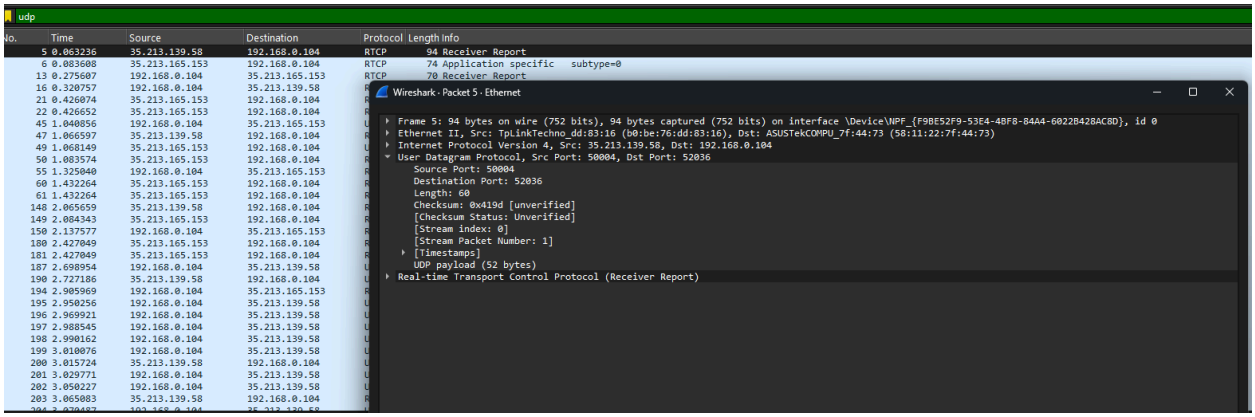
2. รพินทร์ นะราช 6609650624

Wireshark Lab 03 -- UDP

Assignment

1. Select the first UDP segment in your trace. What is the packet number4 of this segment in the trace file? What type of application-layer payload or protocol message is being carried in this UDP segment? Look at the details of this packet in Wireshark. How many fields there are in the UDP header? (You shouldn't look in the textbook! Answer these questions directly from what you observe in the packet trace.) What are the names of these fields?

ANS



1) Packet Number: 5 เป็น UDP Segment แรก ที่ขึ้นมาใน Trace File

2) Application-layer Payload หรือ Protocol: Real-time Transport Control Protocol (RTCP)

3) UDP Header : 4 fields ประกอบด้วย

Source Port : 50004

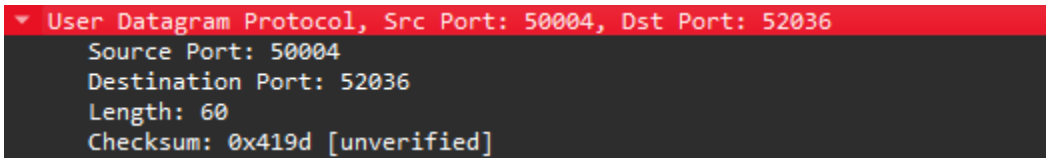
Destination Port : 52036

Length : 60 bytes

Checksum : 0x419d [unverified]

2. By consulting the displayed information in Wireshark's packet content field for this packet (or by consulting the textbook), what is the length (in bytes) of each of the UDP header fields?

ANS



แต่ละฟิลด์จะมีขนาด 2 bytes ตามมาตรฐานของ UDP

Source Port : 50004

Destination Port : 52036

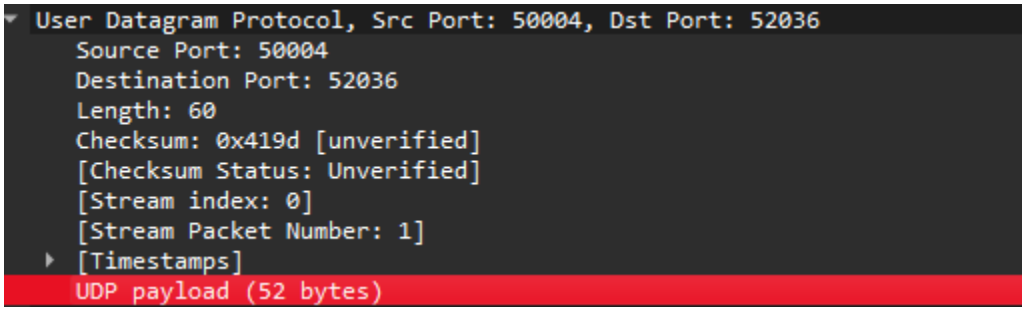
Length : 60 bytes

Checksum : 0x419d [unverified]

รวมเป็น 8 bytes

3. The value in the Length field is the length of what? (You can consult the text for this answer). Verify your claim with your captured UDP packet.

ANS



ค่าของ Length = 60 bytes

UDP Header = 8 bytes

UDP Payload = 60 - 8 = 52 bytes

4. What is the maximum number of bytes that can be included in a UDP payload? (Hint: the answer to this question can be determined by your answer to 2. above)

   ANS

   Maximum IP Packet Size: (2^16 – 1) = 65535 bytes

   ขนาดของ UDP Header คือ 8 bytes

   UDP Payload สูงสุด = 65,535 – 8 bytes = 65527 bytes.

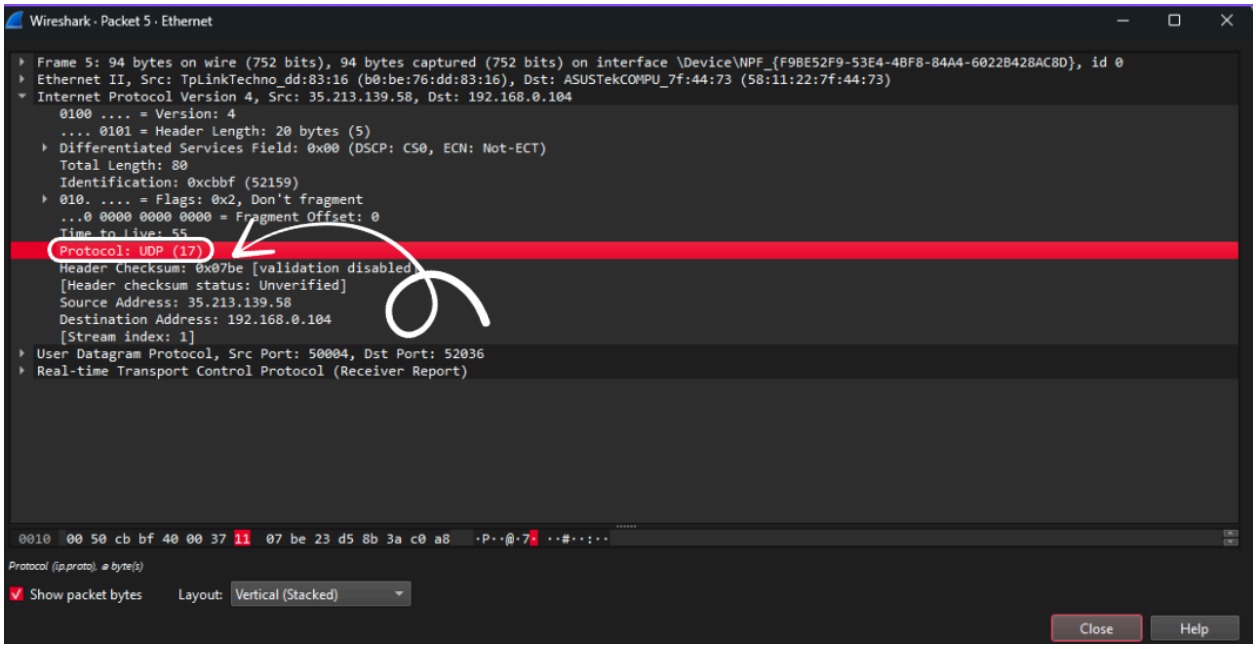
5. What is the largest possible source port number? (Hint: see the hint in 4)

   ANS

   ค่าของที่เป็นไปได้มากสุดของ Maximum IP Packet Size: (2^16 – 1) = 65535 bytes

6. What is the protocol number for UDP? Give your answer in decimal notation. To answer this question, you'll need to look into the Protocol field of the IP datagram containing this UDP segment (see Figure 4.13 in the text, and the discussion of IP header fields)

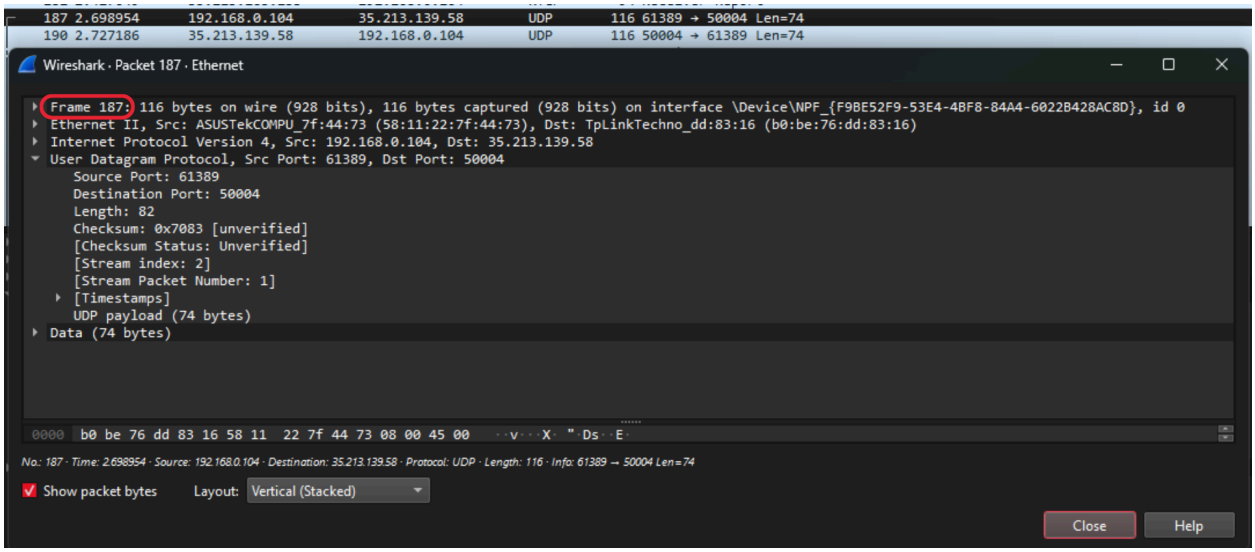
   ANS

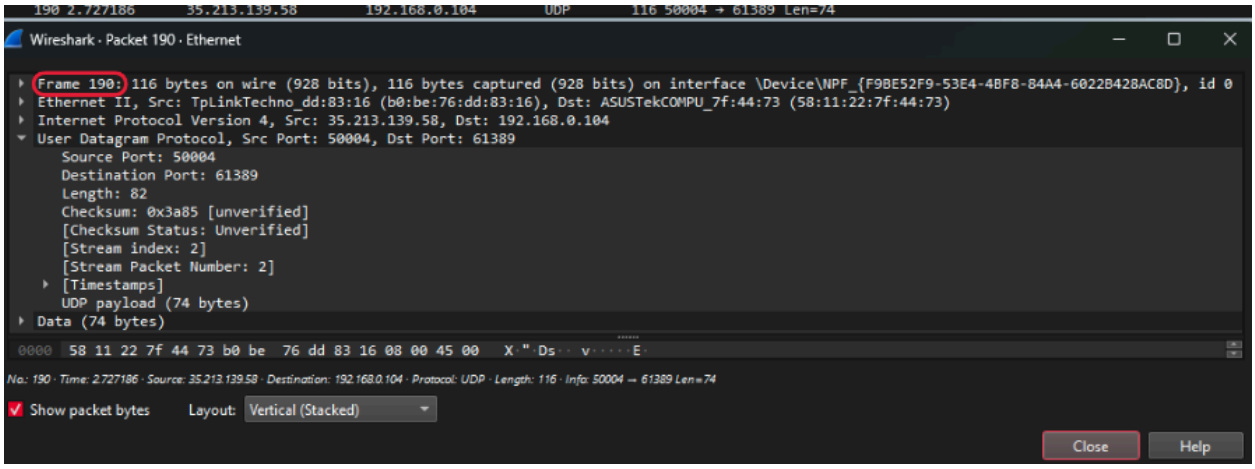


Protocol Number : UDP (17)

7. Examine the pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet. (Hint: for a second packet to be sent in response to a first packet, the sender of the first packet should be the destination of the second packet). What is the packet number5 of the first of these two UDP segments in the trace file? What is the packet number6 of the second of these two UDP segments in the trace file? Describe the relationship between the port numbers in the two packets.

ANS





แพ็กเก็ตแรกที่ส่งออกไป (Request):

Packet Number:187

Source IP: 192.168.0.104

Destination IP: 35.213.139.58

Source Port: 61389

Destination Port: 50004

Length: 82 bytes

Data: 74 bytes

แพ็กเก็ตสองที่ตอบกลับ (Response):

Packet Number: 190

Source IP: 35.213.139.58

Destination IP: 192.168.0.104

Source Port: 50004

Destination Port: 61389

Length: 82 bytes

Data: 74 bytes

พอร์ตปลายทางของแพ็กเก็ตแรก กลายเป็น พอร์ตต้นทางของแพ็กเก็ตที่ตอบกลับ (50004 ↔ 61389)

แสดงว่า เครื่องปลายทาง (35.213.139.58) ตอบกลับมาทางพอร์ตที่เครื่องของเราส่งออกไปในตอนแรก